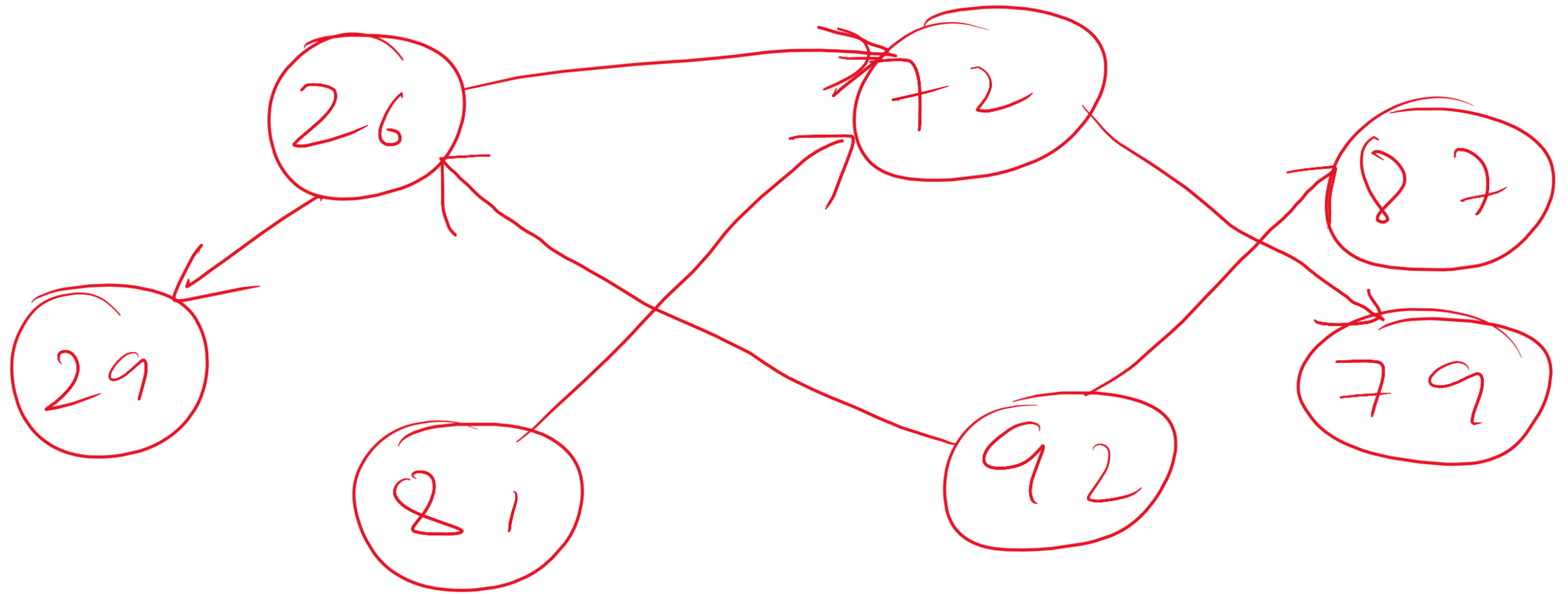


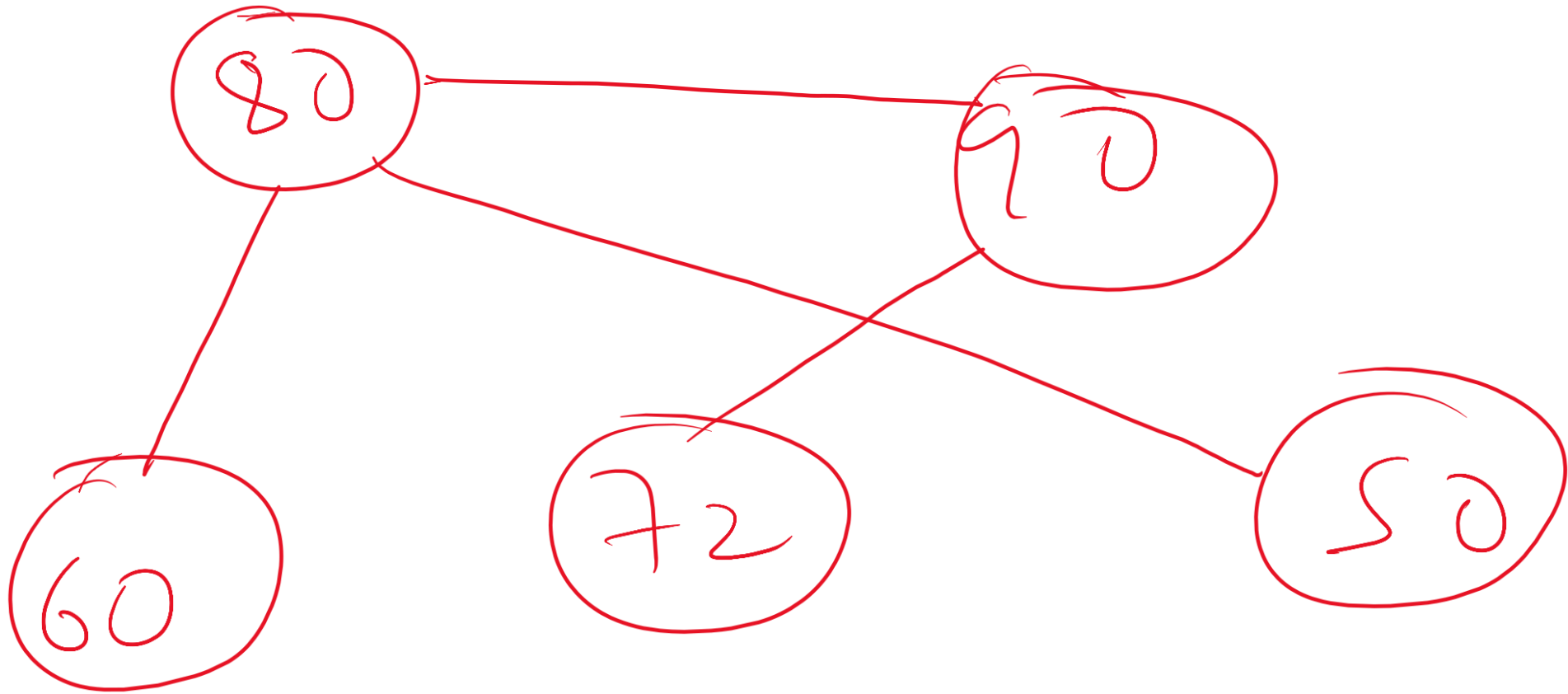
① Graph :- It is a non linear data structure which consists of Nodes (vertices) and Edges where edges are used to connect the vertices. There are 2 types of Graph :-

1. Directed Graph
2. Undirected Graph.

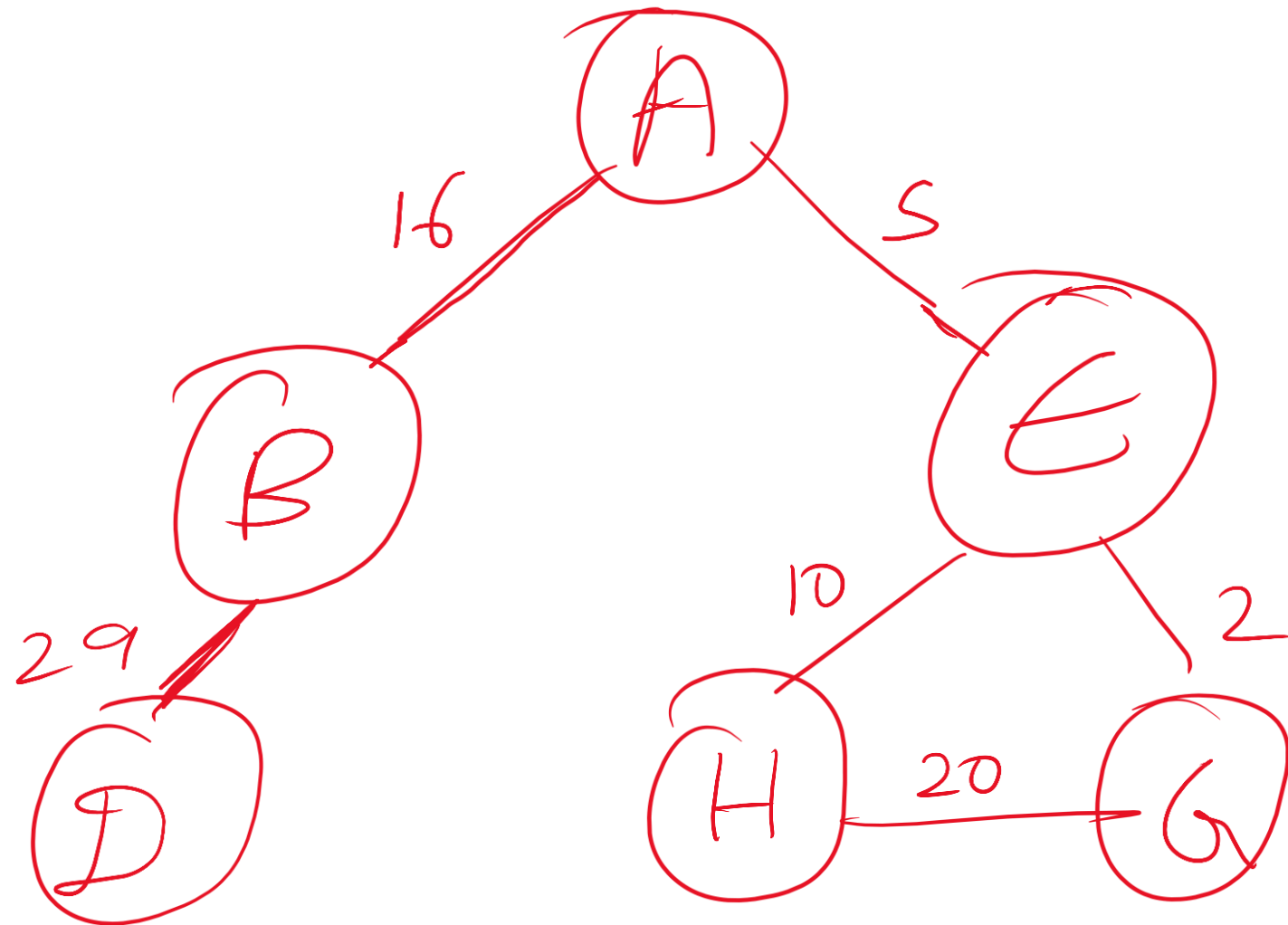
○ Directed Graph :-



① Undirected Graph :-



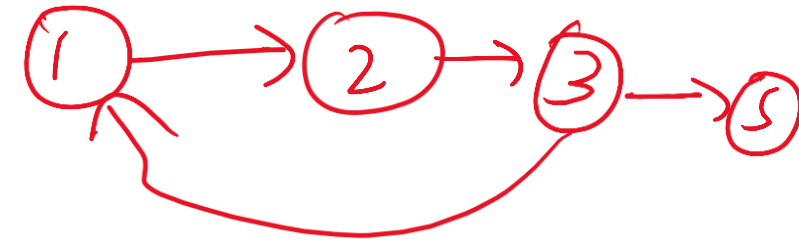
① Weighted Graph :-



① Representation of Graph :-

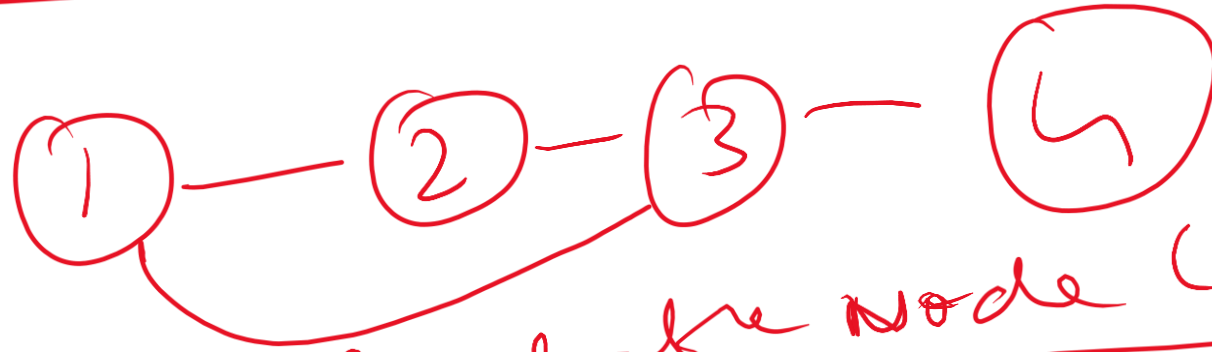
1. Adjacency matrix

2. Adjacency List

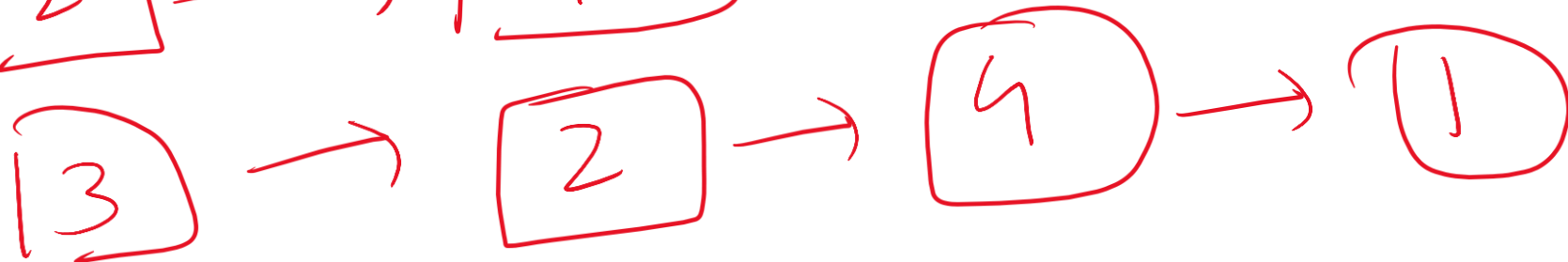
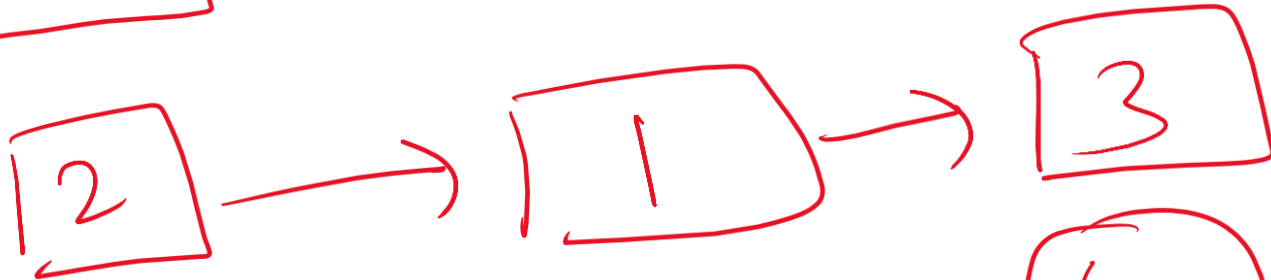
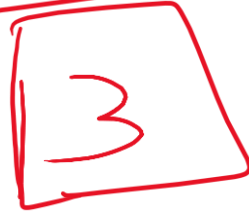
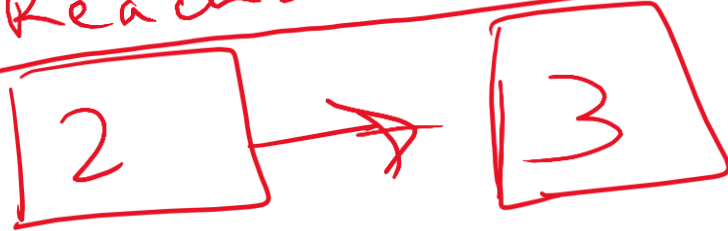


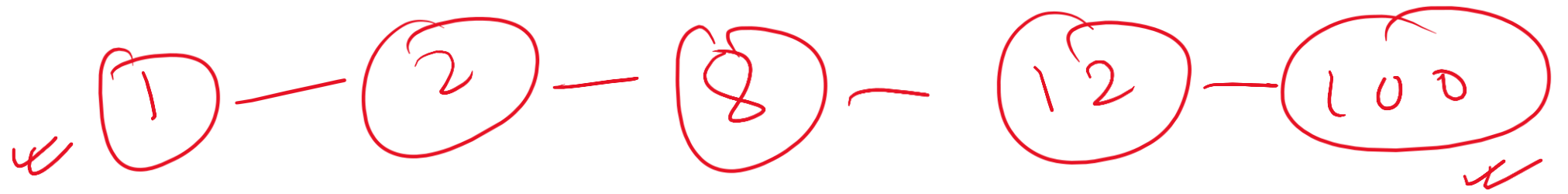
	1	2	3	4	5
1	1	1	0	0	0
2	0	1	1	0	0
3	1	0	1	0	1
4	0	0	0	0	0
5	0	0	0	0	1

① Adjacency List :-



Reachable node (List)

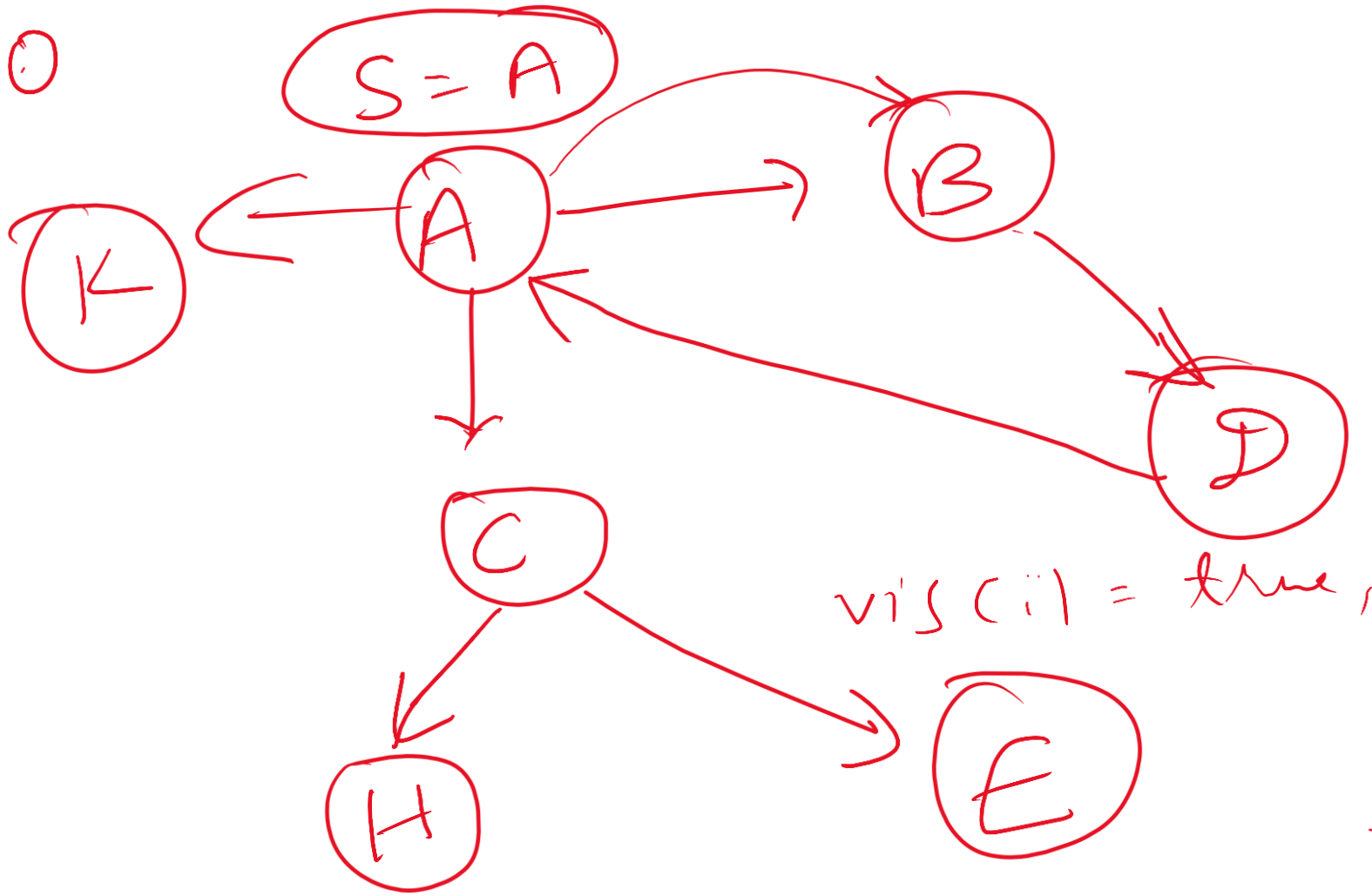




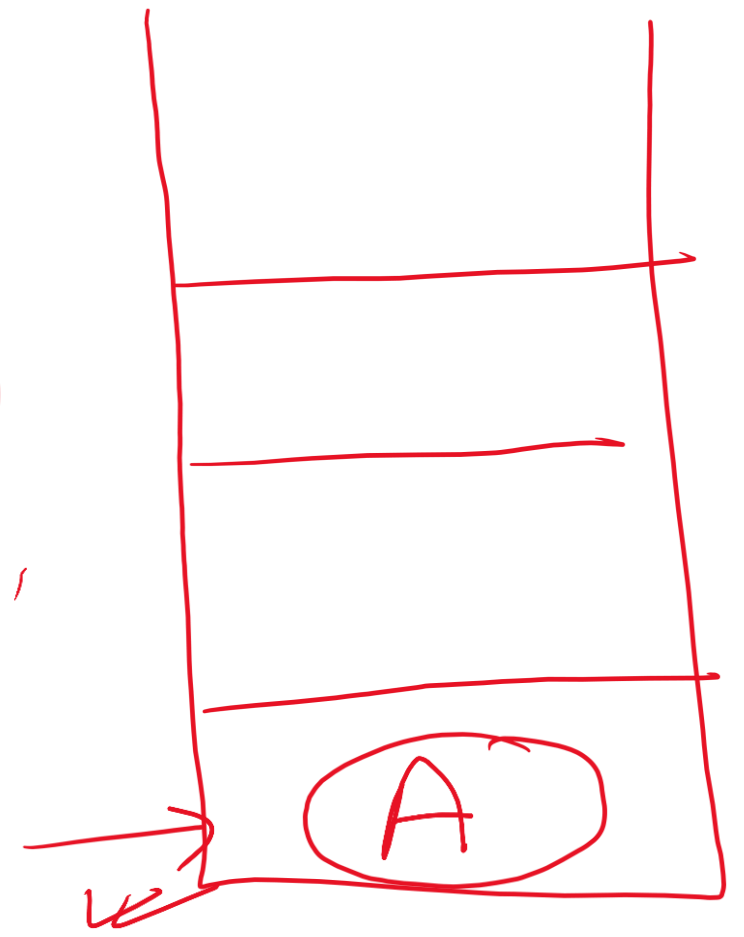
100 × 100 →

① BFS, DFS, Backtracking,
Helper DS, all those tricks
will be applicable in
majority of the problem in
graph DS.

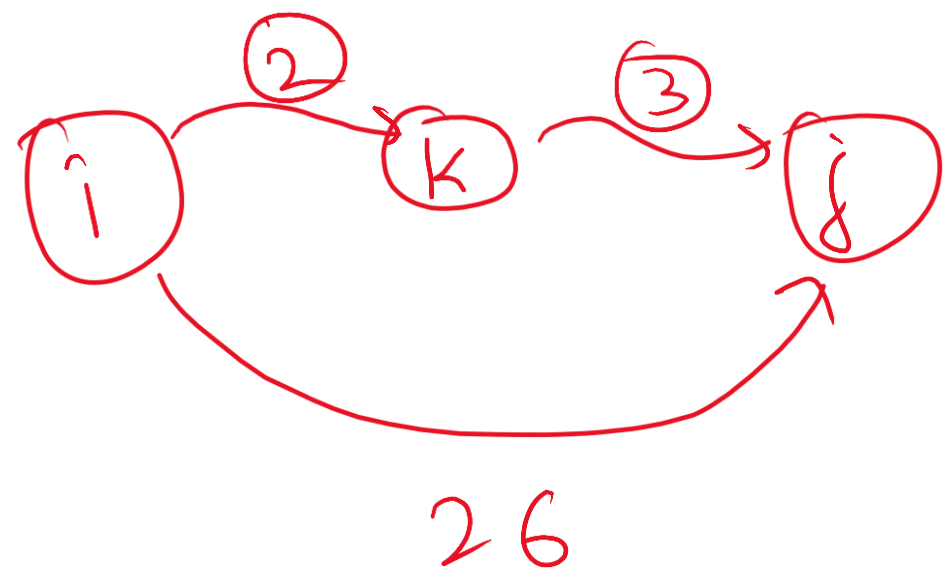
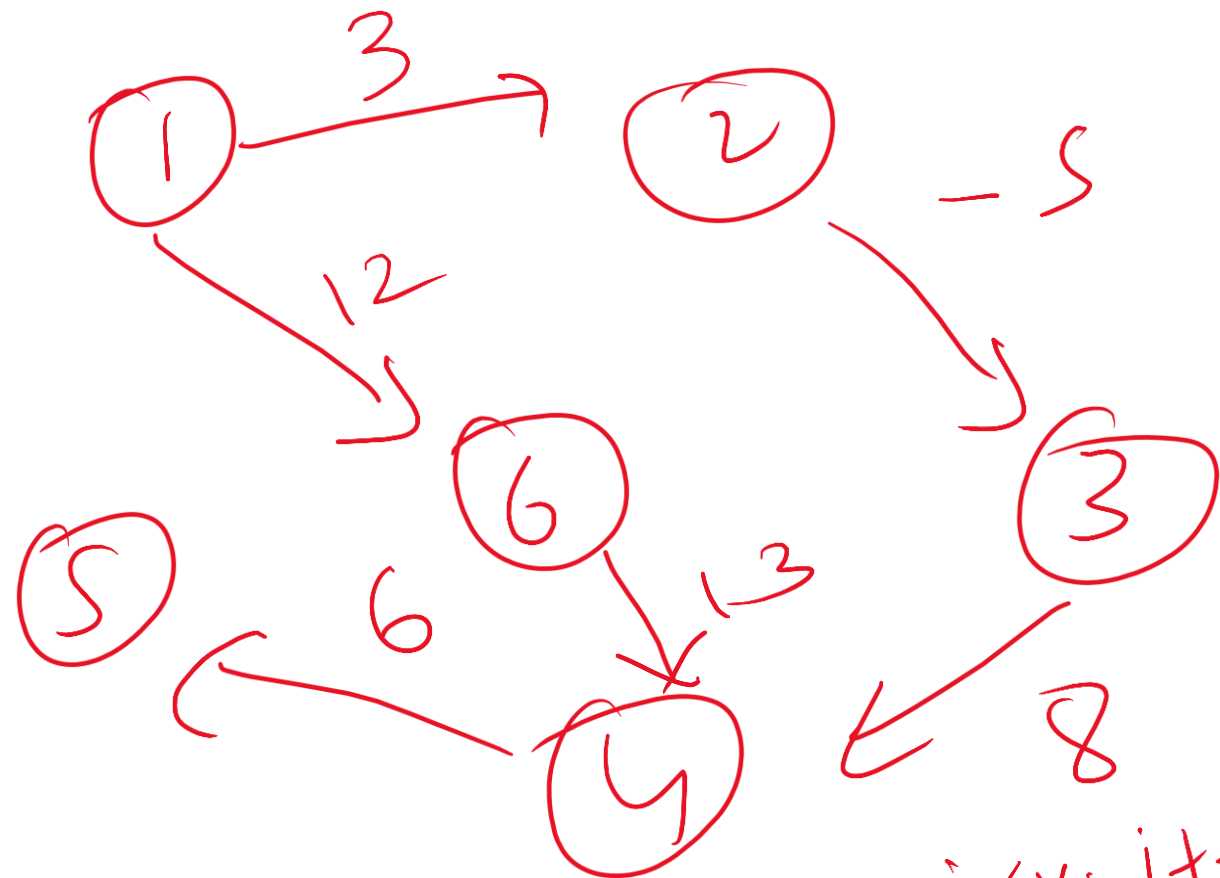
① A binary tree can't
have a loop but a
graph $G = S$ can have
a loop.



$vis(i) = true,$



$recstack(i) = true;$ Rec Stack
→ $recstack(i) = false;$



$$a(i)(j) = \infty$$

for $i = 0, i < v; i++$ $a(i)(k) + a(k)(j)$
 $< a(i)(j)$
 $a(i)(j) = a(i)(k) + a(k)(j);$
 return false;
 return true;